MECA 445	Advanced Programming for AI and Robotics	3	MECA 444L
MECA 595A	BE Summative Learning Project I	1	ENGL 217; Senior Standing
MECH 422	Manufacturing Processes	3	MECH 320
	Engineering Technical Elective II	3	Per course requirements
	Engineering Technical Elective III	3	Per course requirements
Year 4, Spring Semester (18 Credits)			
MECA 542	Industrial and Manufacturing Control	3 (2+2)	ELEC 320
MECA 544	Robotics	3	MECH 321
MECA 595B	BE Summative Learning Project II	3	MECA 595A
	Engineering Technical Elective IV	3	Per course requirements
	Engineering Technical Elective V (AI-Restricted**)	3	Per course requirements
	Social Sciences/ Humanities Elective II	3	
Total BE credits: 152			

* BS-bound students are required to take the MECA 400 – Summative Learning (BS) Project instead.

** Students are required to choose at least one technical elective (i.e., 3 credits) from the list of AI-Restricted Elective courses.

Note: Engineering Technical Electives (levels 400 or above) are selected as such:
9 credit restricted Departmental Electives
6 credits from any engineering discipline at the 400-level or above.

Courses Description

I. Mandatory Courses

Non-Major Courses

Description of the mandatory, non-major courses follows.

CCEE 214	**Programming & Lab**	**3(2,2)**
Understanding of the basic principles of programming and computer systems; Applications of programming to the solution of engineering problems; Control structures, functions, arrays, pointers, and structures. Laboratory experiments. **Prerequisite**: None.		

CCEE 221	**Logic Design**	**3(3,0)**
This course addresses Boolean algebra and logic simplification techniques, data representation, and the design of combinational logic networks for decoders, encoders, multiplexers, and demultiplexers. Design of sequential logic devices,		

including flip-flops, registers, and counters, as well as analysis of devices used to build logic networks, including programmable logic devices. **Prerequisite**: None.

CCEE 221L	Logic Design Lab	1(0,2)
This Lab covers design techniques and implementation of combinational and sequential logic circuits. Experiments include logic gates; the design and implementation of logic circuits; combinational logic circuits (decoders, encoders, multiplexers, demultiplexers, and adders); and the design of sequential logic devices using flip-flops, registers, and counters. **Prerequisite**: CCEE 221. Equivalent to COSC 351L.		

CHEM 211	Environmental Chemistry and Lab	3(2,2)
This course provides an opportunity to develop an understanding of several basic environmental functions, the complicated nature of environmental systems, and the need for multidisciplinary solutions to environmental problems. Topics covered include the hydrosphere, water quantity and quality, soil and the soil ecosystem, biological resources, waste disposal, air pollution, ozone depletion, acid rain, and global warming. The course also includes a set of experiments that provide students with practical experience in various environmental analysis settings, including air and water quality. **Prerequisite**: None.		

CIVE 211	Statics	3(3,0)
Static laws; force vectors and operations; force systems and moments; free-body diagrams; equilibrium of particles and rigid bodies in 2D and 3D; plane trusses and frames; cables; internal forces: shear and moment diagrams; centroid and center of mass; moment of inertia; and friction. **Prerequisite**: None. Annually.		

ELEC 320	Electronics	3(3,0)
This course covers semiconductors, PN junctions, diode theory and circuits, bipolar junction transistors, transistor fundamentals, transistor biasing, amplifiers, MOSFETs, and operational amplifiers. **Prerequisite:** ELEC 210.		

ELEC 320L	Electronics Lab	1(0,2)
This is a lab course that experiments with electrical and electronic circuits. It covers passive electrical elements and sources, lab instruments, voltage divider circuits, Thevenin's theorem, RC circuits, diode circuits, Op-Amp circuits, and BJT and MOSFET characteristics. **Co-requisite:** ELEC 320.		

ELEC 210	Electric Circuits	3(3,0)
This course covers circuit elements and laws; mesh and node equations; network theorems; energy-storage elements; RC, R, and RLC circuits; transformers; sinusoidal (AC) steady-state analysis; power calculation; and an introduction to three-phase circuits. **Prerequisite**: PHYS 191 or Equivalent.		

ELEC 331	Electric Machines	3(3,0)
This course covers magnetic circuits; fundamentals of electromechanical energy conversion; and the construction, testing, steady-state analysis, and operational characteristics of DC machines, transformers, induction machines, and synchronous machines. **Prerequisite**: ELEC 210.		

ENGL 210	English Composition and Rhetoric	3(3,0)
This course reviews the fundamentals of good academic writing in English, teaches essay writing essentials and research skills in two rhetorical modes (persuasion and argumentation), and provides practice in writing essays in these modes, research papers, and oral presentations. **Prerequisite:** ENGL 101 or TOEFL 550+ (paper) or 80+ (computer).		

ENGL 217	Professional English Communication	3(3,0)
This course is designed to help students develop effective professional communication skills, both orally and in writing. In this course, students learn how to write emails, memos, letters, proposals, reports, and other forms of employment correspondence. In addition, this course helps students sharpen their presentation skills. Broadly, this course enables students to behave professionally and effectively in their prospective jobs. **Prerequisite**: ENGL 210.		

ENGR 300	Engineering Economics and Management	3(3,0)
Concepts and techniques in basic Engineering economy principles and applications. Interest and financial mathematics; present worth, annual worth, benefit/cost ratio, internal rate of return, multiple alternatives, income tax, inflation, Risk analysis, Investment and investment choice, equivalence, loans, cost of capital, retirement and replacement, sector analysis and viewpoint, sensitivity analysis, accounting and financial statements. **Prerequisite**: None. Annually.		

ENGR 510	Engineering Project Management	3(3,0)
The course covers key components of engineering project management including project selection and planning, project time management, cost estimation and pricing, contract and specifications, quality management, engineering ethics, and professional conduct, realizing the impact of engineering solutions in various contexts (global, economic, environmental, societal, etc.), sustainability in engineering designs, human resources consideration, communications, risk management, and procurement management. **Prerequisite:** ENGR 300.		

MATH 215	Calculus III	3(3,0)
This course covers complex numbers, trigonometric functions, inverse trigonometric functions, hyperbolic functions, and integration techniques. The course also covers infinite sequences and series: limits of sequences of numbers, bounded sequences, integral test for series, comparison tests, ratio and root tests, alternating series, absolute and conditional convergence, power series, Taylor and MacLaurin series, and applications of power series. Polar functions, polar coordinates, and graphing of polar		

curves are also covered. In addition, topics from multivariable calculus are introduced: double integrals, applications to double integrals, and double integrals in polar form.

MATH 216	Calculus IV	3(3,0)
Multivariable calculus and vector analysis for three-dimensional problems. Topics include 3-D vectors, dot and cross products; quadratic surfaces; limits, continuity, and partial derivatives; gradient, chain rule, and directional derivatives; tangent planes, differentials, extrema, and saddle points; triple integrals in rectangular, cylindrical, and spherical coordinates with applications to mass and moments; vector fields, work, flux, and Green's Theorem; and an introduction to Fourier series. **Prerequisite:** MATH 215.		

MATH 311	Linear Algebra with Applications	3(3,0)
Systems of linear equations, matrix algebra, linear transformations, determinants, vector spaces, eigenvalues and eigenvectors, symmetric matrices, orthogonality, diagonalization.		

MATH 314	Ordinary Differential Equations	3(3,0)
First-order linear differential equations; linear differential equations of second and higher order; linear differential equations with variable coefficients; series solutions; systems of differential equations; Laplace transforms and their inverses. **Prerequisite:** MATH 216.		

MATH 317	Partial Differential Equations	3(3,0)
Lagrange theorem, boundary conditions of first-order equations, non-linear first-order PDEs, Charpit's equations, second-order PDEs, classification: hyperbolic, parabolic, and elliptic, the method of separation of variables**,** introduction to Fourier series and integrals**,** boundary value problems: heat equation, wave equation, Laplace equation, and finite-length strings. **Prerequisite**: MATH 314.		

MATH 351	Probability and Statistics	3(3,0)
Probability and conditional probability; discrete and continuous random variables; marginal distributions; expectation, variance, mean, median, covariance, and correlation; conditional expectation; Normal distribution; sampling distribution; prediction and confidence intervals; hypothesis testing; and regression line and correlation coefficients. **Prerequisite**: MATH 215		

MATH 421	Numerical Analysis	3(3,0)
Error Analysis, solutions of nonlinear equations using fixed-point, Newton-Raphson-Muller's methods, solution of linear systems using Gaussian elimination-iterative methods, interpolation and approximation using Taylor series-Lagrange approximation-Newton polynomials, numerical differentiation and integration, numerical optimization, solutions of ordinary and partial differential equations using Euler's and Heun's and Rung-Kutta methods, MATLAB codes related to the above-mentioned topics. **Prerequisite**: MATH 311.		

MECH 200	Introduction to Mechanical Engineering	3(3,0)
This course aims to provide students with a comprehensive overview of the fundamentals of Mechanical Engineering concepts and applications. Engineering topics to be covered shall include an introduction to learn-by-practice concepts; microcontrollers (Arduino); sensors and actuators; materials, mechanisms, codes, and standards; and robotics (LEGO Mindstorms). In addition, students will be introduced to engineering ethics and will learn how to apply the ASME/ABET code of ethics in practical case studies. The course also integrates the development of hands-on skills, teamwork, and communication; exercises and projects engage students in building, implementing, and testing their designs. **Prerequisite**: None. Annually.		

MECH 220	Dynamics	3(3,0)
Brief review of statics; proper use of different coordinate systems: Cartesian and normal/tangential axes; kinematics of a particle: rectilinear, curvilinear, and relative motion; kinetics of a particle: force and acceleration, work and energy, impulse and momentum; planar kinematics of a rigid body: translating and rotating axes, absolute and relative motion; planar kinetics of a rigid body: force and acceleration, work and energy, impulse and momentum. **Prerequisite**: CIVE 211. Annually.		

MECH 224	Mechanical Engineering Graphics and CAD	3(3,0)
Drafting principles; part and assembly drawings; generating 2D and 3D drawings and solid models using AutoCAD; customizing AutoCAD for specific applications; generating 2D drawings using CAD/CAM software; generating parametric, feature-based solids using Autodesk Mechanical Desktop. Introducing mechanical Revit, a Building Information Modeling (BIM) software tailored for the mechanical, electrical, and plumbing (MEP) disciplines. **Prerequisite**: None. Annually.		

MECH 320	Mechanics of Materials	3(3,0)
Mechanical stress, strain, and stress/strain relations under pure tension, compression, pure bending, and pure torsion of circular bars; 1-D thermal strain & stress; combined stresses; basic thin-walled cylindrical and spherical shells under internal pressure; principal stresses; plane stress transformation & Mohr's circles; basic 3-D elastic stress/strain relations; strain measurement using strain gauge technique. **Prerequisite**: CIVE 211. Annually.		

MECH 320L	Mechanics of Materials Lab	1(0,2)
This hands-on laboratory course complements theoretical concepts from Mechanics of Materials (MECH 320) by providing practical experience in the experimental analysis of material behavior under various loading conditions. Students will conduct tests involving axial tension/compression, bending, shear, and torsion to investigate stress-strain relationships, deformation, and failure modes in structural materials. Emphasis is placed on accurate measurement techniques using strain gauges, extensometers, and deflection indicators, as well as data analysis, interpretation, and comparison with theoretical predictions. Through structured experiments and lab reports, students will develop skills in experimental design, data acquisition, critical analysis, and technical communication. **Co-requisite**: MECH 320.		

MECH 321	Mechanisms and Dynamics of Machinery	3(3,0)
Mechanization of motion; kinematic analysis of linkage mechanisms; graphical synthesis of planar mechanisms; synthesis of cam-follower mechanisms (graphical and analytical designs); gear terminology and types of gears used in machines; analysis and synthesis of gear trains; velocity, acceleration, and force analysis of mechanisms. **Prerequisite**: MECH 200 and MECH 220. Annually.		

MECH 333	Thermal Fluid Sciences	3(3,0)
This course imparts thermal-fluid fundamental concepts (including Thermodynamics, Fluid Mechanics, and Heat Transfer) to non-mechanical engineering students. These cover thermodynamic state and properties of pure substances; concepts of work and heat; energy analysis of closed systems (First Law) and simple flow devices; fluid statics; Bernoulli's equation; Conservation of mass, momentum & energy; viscous flow in pipes; introduction to conduction, convection and radiation heat transfer. **Prerequisite**: MATH 216. Annually.		

MECH 421	Mechanical Design I	3(3,0)
Engineering materials properties; tools of machine design; stress, strain, and deformation of machine parts; buckling of columns; statistical methods and safety consideration; static and fatigue failure theories; design of machine parts for strength under steady loads; design of machine parts for rigidity, deflection of beams and rotation of shafts; design of machine parts for strength under cyclic/dynamic loads; codes and standards. **Prerequisite**: MECH 320. Annually.		

MECH 422	Manufacturing Processes	3(3,0)
General overview of manufacturing; properties of engineering materials; steelmaking; dimensions and tolerances; manufacturing processes: rolling, forging, extrusion, powder metallurgy; mechanical assembly: welding, brazing, soldering, bonding; welding QA/QC; non-destructive testing (NDT); machining; casting; sheet metal forming; plastic injection molding; polymers and composites. **Prerequisite**: MECH 320. Annually.		

PHYS 210	Fundamentals of Physics and Lab	3(3,1)
Welcome to the Introductory Physics course. This course consists of a review of Classical Mechanics, Fluid Mechanics, General Properties of Waves, Electric Forces, Electric Fields for Discrete and Continuous Charge Distributions, Gauss's Law, Electric Potential, Kirchhoff's Rules, Magnetic Fields and Forces, and a Qualitative Discussion of Maxwell's Equations. **Prerequisite**: None.		

Major Courses

Mandatory mechatronics courses are described below.

MECA 340	Dynamic Systems Modeling	3(3,0)
Introduction to dynamic modeling of mechanical, electrical, electromechanical, thermal, fluid, and mixed-discipline systems; Block diagrams and computer simulation (MATLAB); State-variable solution; Laplace Transform solution; Laplace transform		

properties; inverse Laplace transform; transfer function, frequency response, first- and second-order systems. **Prerequisite**: MATH 314. Annually.

MECA 341	Measurements and Instrumentation	3(3,0)
This course covers the fundamentals of instrumentation and measurement of various physical quantities. Topics include sensor types, technologies, characteristics, and calibration; design of a measurement system; statistical analysis of measured data; measurement noise and uncertainties; data acquisition, data storage and display devices; signal conditioning and interface electronics concepts including filtering, A/D and D/A conversion, amplification, modulation, compensation; applications; **Prerequisite**: ELEC 320. Annually.		

MECA 341L	Measurements and Instrumentation Lab	1(0,2)
The lab experience complements the topics covered in MECA 341. Students learn to use the NI ELVIS platform, LabVIEW programming, and the characteristics of data acquisition systems to build a measurement system, perform data analysis and sensor calibration, and design and build signal-conditioning circuits for various types of measurements. The lab involves a team project to integrate all instrumentation and measurement tools and techniques to build a measurement system and submit a report. **Co-requisite**: MECA 341. Annually.		

MECA 400	Summative Learning (BS) Project	3(3,0)
Team-oriented, project-based experience culminating in the creation of an artifact; milestones include project selection and proposal, a creative solution, a report, a presentation, and a demonstration of the device. **Prerequisite**: ENGL 217 for the last semester of the BS Program. Annually.		

MECA 440	Control Systems Design	3(3,0)
This course focuses on the analysis and design of control systems. Measure control system performance; Transient and steady-state response analyses; Topics include analytical solutions in the time and S domains, root locus, Bode, Nyquist, Lead-Lag controller design, and PID compensator design; Analysis and design of control systems in state space. Control system analysis and design using MATLAB and SIMULINK; set up a control system, understand it, and optimize its performance; **Prerequisites**: MECA 340. Annually.		

MECA 440L	Control Systems Design Lab	1(0,2)
This 1-credit laboratory course provides students with practical experience in the design, analysis, and implementation of linear control systems. Students will gain hands-on proficiency in using MATLAB and Simulink as powerful tools for modeling, simulating, and analyzing control system behavior in both the time and frequency domains. A significant component of the lab involves the implementation of fundamental automatic controllers, including ON/OFF, Proportional-Derivative (PD), Proportional-Integral (PI), and Proportional-Integral-Derivative (PID) controllers, using Arduino microcontrollers for various real-world applications such as temperature control and speed regulation. Students will also be introduced to the diverse range of		